THE RELATIVE CYCLE PROPERTY DESCENDS ALONG JOINTLY SURJECTIVE FAMILIES OF FLAT MORPHISMS

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Allowed standard prerequisites (editorial).

- Functoriality of relative cycles under pullback (relative-cycles chapter, Lemma lemma-relative-cycle-functoriality, canonical tag 0H51). cited at proof lines 1033-1034. Flat pullbacks of relative cycles are relative cycles, proving the forward direction.
- Compatibility of specialization maps with flat pullback (relative-cycles chapter, Lemma lemma-specialization-flat-pullback, canonical tag 0H4L). cited at proof lines 1052-1053. Justifies the first equality f_{i,0}^* sp_{X/S}(alpha_eta) = sp_{X_i/S}(f_{i,eta}^* alpha_eta).

Source reference. Original lemma 0H51: sources/stacks-chow/tags.txt, source lines 21371–21371. Cited at proof lines 1033-1034: forward direction.

Source reference. Original lemma 0H4L: sources/stacks-chow/tags.txt, source lines 21356–21356. Cited at proof lines 1052-1053: the first pullback equality.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume S locally Noetherian and f locally of finite type. Let $r, e \geq 0$ be integers. Let α be a family of r-cycles on fibres of X/S. Let $\{f_i : X_i \to X\}$ be a jointly surjective family of flat morphisms, locally of finite type, and of relative dimension e. Then α is a relative r-cycle if and only if each flat pullback $f_i^*\alpha$ is a relative r-cycle.*

Proof. If α is a relative r-cycle, then each pull back $f_i^*\alpha$ is a relative r-cycle by Lemma 0H51. Assume each $f_i^*\alpha$ is a relative r-cycle. Let $g : S' \to S$ be a morphism where S' is the spectrum of a discrete valuation ring. After replacing S by S', X by $X' = X \times_S S'$, and α by $\alpha' = g^*\alpha$ we reduce to the problem studied in the next paragraph.

Assume S is the spectrum of a discrete valuation ring with closed point 0 and generic point η. We have to show that $sp_{X/S}(\alpha_\eta) = \alpha_0$. Denote $f_{i,0} : X_{i,0} \to X_0$ the base change of f_i to the closed point of S. Similarly for $f_{i,\eta}$. Observe that

$$f_{i,0}^* sp_{X/S}(\alpha_\eta) = sp_{X_i/S}(f_{i,\eta}^*\alpha_\eta) = f_{i,0}^*\alpha_0$$

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Namely, the first equality holds by Lemma 0H4L and the second by assumption. Since the family of maps $f_{i,0}^* : Z_r(X_0) \to Z_r(X_{i,0})$ is jointly injective (due to the fact that $f_{i,0}$ is jointly surjective), we conclude what we want. $\square$